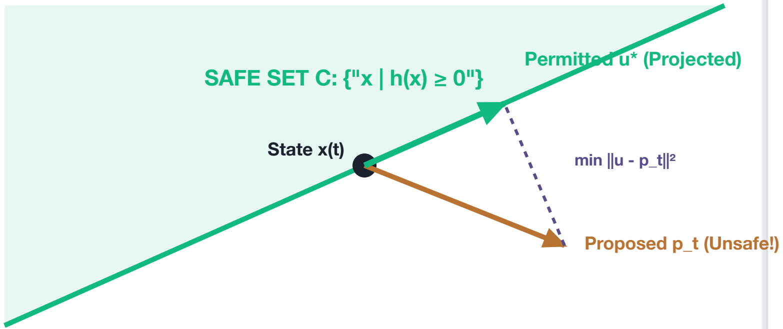
CONTROL BARRIER FUNCTION (CBF) MINIMAL-INTERVENTION PROJECTION

Quadratic Program Filter: min ||u - p_t||² subject to L_f h(x) + L_g h(x) u + α(h(x)) ≥ 0



THE BARRIER GUARANTEE

- **Minimal Intervention Principle:**
  If proposed action p_t is safe, u* = p_t with zero distortion.
  If p_t violates safety, QP applies minimal correction.
- **Forward Invariance (Nagumo's Theorem):**
  If state starts in C ($h(x_0) \geq 0$), it remains inside C
  for all future time t ≥ 0: h(x(t)) ≥ 0 certified.
- **Deterministic Active-Set QP Solver:**
  Formulated with linear constraints in 3–6 variables.
  Guaranteed convergence in ≤ 15 iterations (≤ 175 μs).
- **Zero Dynamic Allocation (malloc=0):**
  Matrices pre-allocated in static SRAM.
  Zero page faults or heap jitter in real-time loop.